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B.Tech. Degree VI Semester Examination April 2019

ME 15 -1602 MACHINE DESIGN I (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I.
 - (a) What is meant by stress concentration?
 - (b) How do the size and surface condition of a component affect the strength?
 - (c) How do you consider creep stress in design?
 - (d) What is self-locking of power screw?
 - (e) Give any two practical applications of couplings?
 - (f) What is meant by preloading of bolts?
 - (g) What are the modes of failure of riveted joints?
 - (h) What do you mean by surge in springs?
 - (i) Write any two advantages of welded joints compared to riveted joints.
 - (j) What are the merits to prefer hollow shaft over solid shaft?

PART B

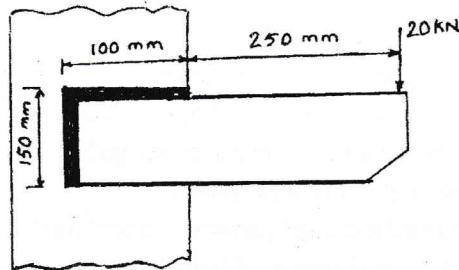
(4 × 10 = 40)

- II. State and explain the theories of failure in design. (10)
- OR**
- III. A rod of circular cross section is subjected to an alternating tensile force varying from 25 kN to 75 kN. Determine the diameter of rod according to Gerber methods, Goodman method and Soderberg method using the following material properties. Take ultimate strengths as 1100 MPa, Yield strength as 600 MPa, Factor of safety as 2 and neglect other correction factors. (10)
- OR**
- IV. A double thread power screw, used for lifting the load, has a nominal diameter of 30 mm and a pitch of 6 mm. The coefficient of friction at the screw threads is 0.1. Neglecting collar friction, calculate efficiency of the screw with square threads and Acme threads. (10)
- OR**
- V. Design a socket and spigot type cotter joint to connect two rods of diameter 30 mm. The following stresses may be used. Tensile stress = 90 MPa, Shear stress = 60 MPa and Crushing stress = 150 MPa. (10)
- OR**
- VI. An engine indicator has a plunger diameter of 20 mm. When the steam pressure on the plunger is 5 MPa, the indicator spring is compressed by 12 mm. For permissible shear stress of 450 MPa, spring index of 5 and modulus of rigidity 80×10^3 MPa, design the helical compression spring. (10)

OR**(P.T.O.)**

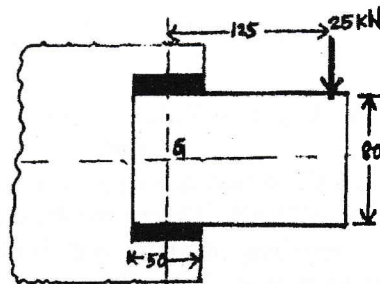
- VII. Design a double riveted butt joint with two cover plates for the longitudinal seam of boiler shell 1.5 m in diameter subjected to a steam pressure of 0.95 N/mm^2 . Assume joint efficiency as 75%, allowable tensile stress in the plate 90 MPa, compressive stress 140 MPa and shear stress in the rivet 56 MPa. (10)

- VIII. (a) What are the stresses to be considered while designing a shaft for transmitting power? (3)
 (b) Determine the size of weld for a plate welded as shown in the figure. The allowable stress for the weld is 75 MPa. (7)



OR

- IX. (a) What is meant by critical speed of shafts? Explain. (3)
 (b) A bracket carrying a load of 25 kN is to be welded as shown in the figure. Find the size of weld required if the allowable shear stress is not to exceed 90 MPa. (7)



(All dimensions in mm)
